

Decision and Social Choice

PE3101P — Lecture 7: Harsanyi's aggregation theorem

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1st October, 2026

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 - ▶ **Weeks 11–12:** voting theory; the alleged irrationality of voting.

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- ▶ **Optional make-up tutorial** (the Friday tutorial falls on Well-Being Day): Thursday 8 October, 10:00–12:00, AS3-0305.

Today

- 1 Social Choice
- 2 Aggregation
- 3 The Proof
- 4 Objections
- 5 Utility and Wellbeing
- 6 More on the Premises

Choosing for One, Choosing for Many

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- ▶ But that doesn't tell us how to choose for several people, when the decision affects them differently.
- ▶ They can disagree, so there might be no option that is "best for everyone".
- ▶ We need a way to balance people's interests against each other. Harsanyi's aggregation theorem gives an argument for one way of doing that.

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- ▶ We're interested in cases where people are affected differently: one is made better off, the other worse off.
- ▶ So we judge everything from *one* set of credences: the same probabilities for everyone.

A Bigger Benefit, or a Smaller One?

	Option 1	Option 2
Alex	8	2
Darius	3	5

Wellbeing levels.

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- ▶ One natural answer: benefit someone by more, rather than someone else by less. Choose Option 1.

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Question

Could we get numbers like these from their preferences alone, using their utility functions? What might go wrong?

Stretching a Utility Function

If u represents someone's preferences,
so does $au + c$, for any $a > 0$.

	Option 1	Option 2
Alex's utility	8	2
Darius's utility	3	5

The same numbers, now read as utilities representing their preferences. Alex's differ by 6; Darius's by 2.

Stretching a Utility Function

If u represents someone's preferences,
so does $au + c$, for any $a > 0$.

	Option 1	Option 2
Alex's utility	8	2
Darius's utility	30	50

Stretch Darius's utility function by a factor of 10. It represents exactly the same preferences, but now his difference is 20.

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- ▶ So if all we look at is their preferences, there's no way to tell which option makes more of a difference.
- ▶ We have to bring in some other information. That's the main reason we talk about wellbeing.

We Need Comparisons of Wellbeing

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Darius	4	5

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- ▶ Preferences are enough for decision theory, but arguably not to decide what's best overall.

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- ▶ Overall, the best person to benefit is Alex. But unless we can compare their wellbeing, we can't say that.
- ▶ Preferences are enough for decision theory, but arguably not to decide what's best overall.
- ▶ (Deciding from preferences alone is still sometimes useful: that's voting theory, later in the semester.)

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Not: X is at least as *preferred* by i as Y .

But: X is at least as *good for* i as Y . Written $X \succsim_i Y$.

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- ▶ It ranks prospects, not just outcomes: it takes account of each person’s risks and benefits.
- ▶ An outcome counts as a prospect: the one that gives that outcome for sure.

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 - ▶ which outcome *ought to be chosen*, morally;
 - ▶ which outcome would be *morally best*.
- ▶ I'll mostly have the last in mind: which outcome is best, when the choice involves trade-offs between people.

Harsanyi's Aggregation Theorem

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Utilitarianism

The best outcome is the one with the greatest sum of wellbeing*:

$$X \succsim Y \quad \text{iff} \quad w_1(X) + \cdots + w_n(X) \geq w_1(Y) + \cdots + w_n(Y),$$

where $w_i(X)$ is person i 's wellbeing in outcome X .

*Strictly, a sum of utilities chosen to match wellbeing: more after the proof.

Same Total, Equally Good

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

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- Utilitarianism: these are equally good.

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- ▶ Is giving everything to Alex really equally as good as sharing everything out equally?!

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- ▶ Utilitarianism: these are equally good.
- ▶ Is giving everything to Alex really equally as good as sharing everything out equally?!
- ▶ The proof will get from UNEQUAL to EQUAL one step at a time. Watch for the step you would resist.

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- ▶ Writing outcomes as (Alex, Darius): $(10, 0) \sim (0, 10)$.
- ▶ Alex and Darius matter equally, and what's at stake for them is equal. So we should be indifferent about who gets what.
- ▶ With Philip too, as (Alex, Darius, Philip):
 $(1, 2, 3) \sim (2, 3, 1)$. And so on, for any number of people.

Statewise Dominance

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If prospect X is at least as good as prospect Y in every state of nature, then $X \succsim Y$. If X is also better than Y in some state of nature, then $X \succ Y$.

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	Disaster	No disaster
Prepare	less damage	as normal
Don't prepare	more damage	as normal

Preparing costs nothing. If the disaster happens, preparing was better; if not, it's equally good either way.

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- ▶ “Ex ante”: it applies to prospects, not just to outcomes.
- ▶ Choosing for many people still involves risk. Think of economic policy.
- ▶ “Better for everyone” doesn’t mean guaranteed to turn out better for everyone. It means a better balance of risks and benefits for each person.

Expected Utility: What Is Better for Each Person

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- ▶ The numbers we assign are still utilities: just utility functions constrained by the facts about wellbeing.
- ▶ What exactly that means: after the proof.

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If $X \succsim Y$ and $Y \succsim Z$, then $X \succsim Z$.

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- This is what chains the steps of the proof together.

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Also assumed: every distribution of wellbeing, and every lottery over them with finitely many outcomes, is possible.

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They all look fairly innocuous. But together, they imply utilitarianism.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

UNEQUAL

	Heads	Tails
Alex	10	10
Darius	0	0

UNEQUAL for sure, written out with a fair coin:
Alex gets 10 and Darius gets 0, however it lands.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

	Heads	Tails
Alex	10	10
Darius	0	0

Now look at each state of nature separately.
If Tails, the outcome is (10, 0).

A Simple Case (1): UNEQUAL and EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Swap Alex and Darius in Tails: (0, 10).
By Anonymity, this is equally good.

A Simple Case (1): UNEQUAL and EQUAL CHANCES

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Equally good in every state (Heads is unchanged).

By Statewise Dominance, both ways round:

UNEQUAL $\sim$ EQUAL CHANCES.

A Simple Case (2): EQUAL CHANCES and EQUAL

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Now look at each person separately.

A Simple Case (2): EQUAL CHANCES and EQUAL

	Heads	Tails
Alex	10	0
Darius	0	10

Alex's expected wellbeing: $\frac{1}{2} \cdot 10 + \frac{1}{2} \cdot 0 = 5$.

A Simple Case (2): EQUAL CHANCES and EQUAL

	Heads	Tails
Alex	5	5
Darius	0	10

Give Alex 5, however the coin lands.
His expected wellbeing is still 5.

A Simple Case (2): EQUAL CHANCES and EQUAL

	Heads	Tails
Alex	5	5
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Darius's expected wellbeing: $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 10 = 5$.

A Simple Case (2): EQUAL CHANCES and EQUAL

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By Expected Utility, equally good for each person.

By Ex Ante Pareto, both ways round:

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Putting It Together

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(10, 0) is exactly as good as (5, 5). They have the same total.

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That's the heart of the argument. The general case makes the same two moves with any number of people, then uses them to rank any two outcomes.

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- ▶ The average is 2, so $E(X) = (2, 2, 2)$.

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- ▶ Three people, Alex, Darius and Philip: $X = (1, 2, 3)$.
- ▶ The average is 2, so $E(X) = (2, 2, 2)$.
- ▶ For $X = (10, 0)$: $E(X) = (5, 5)$, which is EQUAL.

The Lottery-Equalisation of an Outcome

Lottery-Equalisation

A lottery-equalisation of X , written $LE(X)$ (any one will do): n equally likely states, one per person. In each state, the outcome is a rearrangement of X , and each person takes each place in X in exactly one state.

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$LE(X)$, for $X = (1, 2, 3)$

	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

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Each state has probability $\frac{1}{3}$.

For $X = (10, 0)$: $LE(X)$ is EQUAL CHANCES.

Step 1: $X \sim \text{LE}(X)$

	X , for sure		
	State 1	State 2	State 3
Alex	1	1	1
Darius	2	2	2
Philip	3	3	3

Each state has probability $\frac{1}{3}$.

$X = (1, 2, 3)$ for sure, written out with three equally likely states.

Step 1: $X \sim \text{LE}(X)$

	LE(X)		
	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

In each state, the outcome is a rearrangement of X .
By Anonymity, in every state the outcome of this prospect is
equally as good as X .

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	LE(X)		
	State 1	State 2	State 3
Alex	2	3	1
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Each state has probability $\frac{1}{3}$.

So the two prospects are equally good in every state.
By Statewise Dominance, both ways round: $X \sim \text{LE}(X)$.

Step 2: $LE(X) \sim E(X)$

	LE(X)		
	State 1	State 2	State 3
Alex	2	3	1
Darius	3	1	2
Philip	1	2	3

Each state has probability $\frac{1}{3}$.

Each person's expected wellbeing: $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 2 + \frac{1}{3} \cdot 3 = 2$.

Step 2: $LE(X) \sim E(X)$

E(X)			
	State 1	State 2	State 3
Alex	2	2	2
Darius	2	2	2
Philip	2	2	2

Each state has probability $\frac{1}{3}$.

In $E(X)$, each person gets 2 for sure.

Step 2: $LE(X) \sim E(X)$

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Alex	2	2	2
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By Expected Utility, equally good for each person.
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Step 3: $X \sim E(X)$

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Step 1 (Anonymity, Statewise Dominance)

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$X \sim E(X)$ Transitivity

Every outcome is equally as good as its equalisation.

(In the simple case: $(10, 0) \sim (5, 5)$.)

Step 4: Utilitarianism

- ▶ Compare any two outcomes X and Y .

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- ▶ If the totals are equal, $E(X)$ and $E(Y)$ are the same outcome, so $X \sim Y$.

$X \succsim Y$ iff X has at least as great total wellbeing as Y .

Can We Reject the Conclusion?

	Alex	Darius	Total
UNEQUAL	10	0	10
EQUAL	5	5	10

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UNEQUAL $\sim$ EQUAL CHANCES Statewise Dominance (and Anonymity)

EQUAL CHANCES $\sim$ EQUAL Ex Ante Pareto (and Expected Utility)

Diamond's Objection

	Heads	Tails
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- ▶ But it is no better in either state. If Heads, the outcome is the same. If Tails, $(0, 10) \sim (10, 0)$, by Anonymity.
- ▶ So Statewise Dominance says they're equally good. If the coin toss really is better, Statewise Dominance (also called the sure-thing principle) is not an appropriate assumption for the social welfare ordering.

The Objection to Ex Ante Pareto (1)

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

EQUAL

	Heads	Tails
Alex	5	5
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EQUAL CHANCES			EQUAL		
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- ▶ The other move. Ex Ante Pareto: equally good. Each person's expected wellbeing is 5 either way.
- ▶ But EQUAL CHANCES guarantees inequality: one person is sure to end up worse off than the other.

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- ▶ We might use a lottery to “split” an indivisible good. But when the thing can be shared equally, people usually think the best thing is *to* share it.
- ▶ Intuitively, EQUAL CHANCES is at best an imperfect alternative to sharing.

Decide As You Would with Full Information

The Principle of Full Information

If you know that, whichever state obtains, you would rightly rank X above Y once you knew it, then rank X above Y now.

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- ▶ So EQUAL is better than EQUAL CHANCES, whatever happens. But Ex Ante Pareto says they are equally good.

The Harsanyi–Sen–Weymark Debate

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- ▶ The objection: it shows that overall good is a sum of utilities. Nothing connects utility with wellbeing. So it doesn't show utilitarianism.
- ▶ So does the theorem really show utilitarianism? How might a utilitarian reply to the objection?

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- ▶ By Expected Utility, each person's betterness relation has a utility function: numbers whose expectation says what's better for that person.
- ▶ But only up to stretching and shifting: the problem we started with.
- ▶ So which utility functions are we using? And how do they relate to wellbeing?

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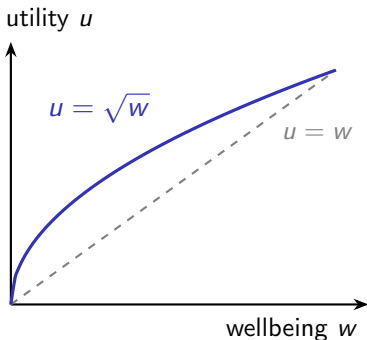
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- ▶ The two can come apart, by a curve.

Utility as a Function of Wellbeing

Two ways utility could curve against wellbeing.
The dashed line, $u = w$, is for comparison.

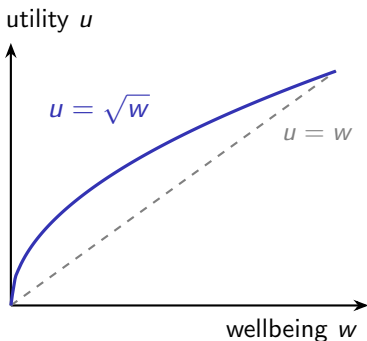


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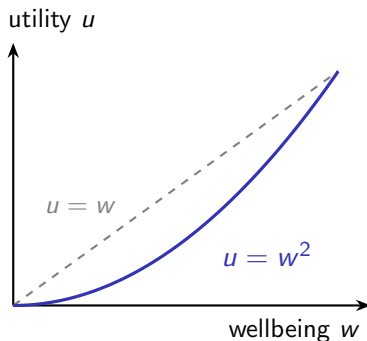
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Convex:
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- ▶ So they're equally good for Darius, though the gamble gives more wellbeing on average (5). His betterness is risk averse about wellbeing.
- ▶ But it's never risk averse about utility: by definition, expected utility is what's better for him.

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- ▶ But not necessarily twice the utility, twice the wellbeing.

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- ▶ Suppose Alex's betterness is risk neutral about wellbeing, and Darius's is risk averse:

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- ▶ Suppose Alex's betterness is risk neutral about wellbeing, and Darius's is risk averse:

Wellbeing	0	1	4	9
Alex's utility = wellbeing	0	1	4	9
Darius's utility = $\sqrt{\text{wellbeing}}$	0	1	2	3

Match them at 0 and 1, and they come apart at 4 and 9. No stretching or shifting fixes that.

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- ▶ But betterness tacitly uses the correct risk attitude. We're judging the best balance of risk for that person, which doesn't seem to depend so much on the attitudes of the people in question.
- ▶ What's good for you is good for you, regardless of what you prefer (arguably, anyway).

What the Theorem Really Says

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- ▶ For each person: the numbers whose *expectation* says what's good for them.
- ▶ The theorem: the *sum* of those same numbers says what's good overall.
- ▶ “The greatest sum of wellbeing” is exactly right only if betterness is risk neutral about wellbeing: utility a straight line in wellbeing.
- ▶ Otherwise, it's the greatest sum of utilities. That's the asterisk.

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- ▶ So the debate is about whether VNM utilitarianism is utilitarianism.

Equally Good, or Incomparable? (1)

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 - ▶ $(10, 0)$ and $(0, 10)$ are *incomparable*: neither better, nor worse, nor equally good.
- ▶ If they're equally good, then $(11, 0) \succ (10, 0) \sim (0, 10)$: Alex getting 11 is better than Darius getting 10.

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- ▶ Anonymity and Statewise Dominance work together: they give UNEQUAL $\sim$ EQUAL CHANCES.
- ▶ If that's the step we find implausible, we could reject Statewise Dominance — or Anonymity.

Rejecting Anonymity Instead of Statewise Dominance

- ▶ Rejecting Anonymity can't mean it's better for one of them to get the 10. So $(10, 0)$ and $(0, 10)$ are incomparable, and Statewise Dominance no longer gives $\text{UNEQUAL} \sim \text{EQUAL CHANCES}$.

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- ▶ The intuition is that EQUAL CHANCES is better than either outcome for sure. So it must be better than both $(10, 0)$ and $(0, 10)$ —
- ▶ — even though, in each state of nature, its outcome is one of those two, and so is no better than them.

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- ▶ Many people find Negative Dominance quite plausible.
- ▶ So perhaps one could avoid the theorem's conclusion by dropping Anonymity. But it's not so clear that that's a good way of doing it.

From Pairwise Anonymity to Anonymity (1)

Pairwise Anonymity

If outcome Y can be obtained from outcome X by swapping the wellbeing levels of two people, then $X \sim Y$.

$(1, 2, 3)$

Alex, Darius and Philip. Pairwise Anonymity only lets us swap two people at a time. Can we still get $(1, 2, 3) \sim (2, 3, 1)$?

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$$(1, 2, 3) \sim (2, 1, 3)$$

Swap Alex and Darius. Equally good, by Pairwise Anonymity.

From Pairwise Anonymity to Anonymity (1)

Pairwise Anonymity

If outcome Y can be obtained from outcome X by swapping the wellbeing levels of two people, then $X \sim Y$.

$$(1, 2, 3) \sim (2, 1, 3) \sim (2, \mathbf{3}, \mathbf{1})$$

Now swap Darius and Philip. Equally good, by Pairwise Anonymity.

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- ▶ Each swap gives an equally good outcome, by Pairwise Anonymity.
- ▶ So, by Transitivity, any rearrangement is equally good: Anonymity.
- ▶ This works for any finite number of people. With infinitely many, it doesn't.

Ex Ante Pareto, One Person at a Time

- ▶ We only need Ex Ante Pareto for changes to one person's prospect: if only i 's prospect changes, and it's equally good for i , the result is equally good.

EQUAL CHANCES

	Heads	Tails
Alex	10	0
Darius	0	10

Start from EQUAL CHANCES.

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- ▶ We only need Ex Ante Pareto for changes to one person's prospect: if only i 's prospect changes, and it's equally good for i , the result is equally good.

	Heads	Tails
Alex	5	5
Darius	0	10

Change only Alex's prospect: 5 for sure. Equally good for him, and nothing changes for Darius. So equally good.

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Alex	5	5
Darius	5	5

Now change only Darius's prospect. Equally good again.

Ex Ante Pareto, One Person at a Time

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EQUAL

	Heads	Tails
Alex	5	5
Darius	5	5

By Transitivity: EQUAL CHANCES $\sim$ EQUAL. The same works in general: one person at a time.

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- ▶ Why: concentrate each outcome's wellbeing in one person, then remove the neutral lives. Next slide.
- ▶ Zero: 0 is the level where a life would be neither good nor bad. That fixes which lives are neutral.

Concentrate, Then Remove the Neutral Lives

A dash: that person doesn't exist.

	Alex	Darius	Philip	Total	Why
X	3	1	2	6	

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~	7	—	—	7	Anonymity across different people

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Y	2	5	—	7	
$\sim$	0	7	—	7	The theorem
$\sim$	—	7	—	7	Neutral Addition
$\sim$	7	—	—	7	Anonymity across different people

$(7, -, -) \succ (6, -, -)$: Alex at 7 is better than Alex at 6.

So, by Transitivity, $Y \succ X$: the greater total wins.

Optional Readings for Next Week

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- ▶ Adler, Matthew D. and Nils Holtug (2019). “Prioritarianism: A Response to Critics”. In: *Politics, Philosophy & Economics* 18.2, pp. 101–144.

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